

Part 4 -- Clustering: Grouping S&P 500 Companies

1. Dataset Description

Source: S&P 500 Stocks (daily updated) -- Kaggle

We combined two files for this unsupervised analysis:

- sp500_stocks.csv: Aggregated per-company stock metrics -- average daily return, volatility, average volume, average close, average daily range.
- sp500_companies.csv: Company fundamentals -- Current Price, Market Cap, EBITDA, Revenue Growth, Full-time Employees, Index Weight.

Total features: 11 numeric features, no target variable (fully unsupervised).

Why this dataset? Grouping S&P 500 companies by financial behavior reveals natural market segments -- mega-cap tech vs. traditional industrials vs. high-growth firms -- useful for portfolio diversification and sector analysis.

2. Methodology

Preprocessing

- Aggregated daily stock data per company (mean return, std of returns as volatility, mean volume, mean close).
- Merged with company fundamentals (Market Cap, EBITDA, Revenue Growth, Employees, Weight).
- Dropped rows with missing values.
- StandardScaler applied to all features before every clustering algorithm.
- PCA reduced to 2 components for visualization only; clustering was performed on the full 11-feature scaled set.

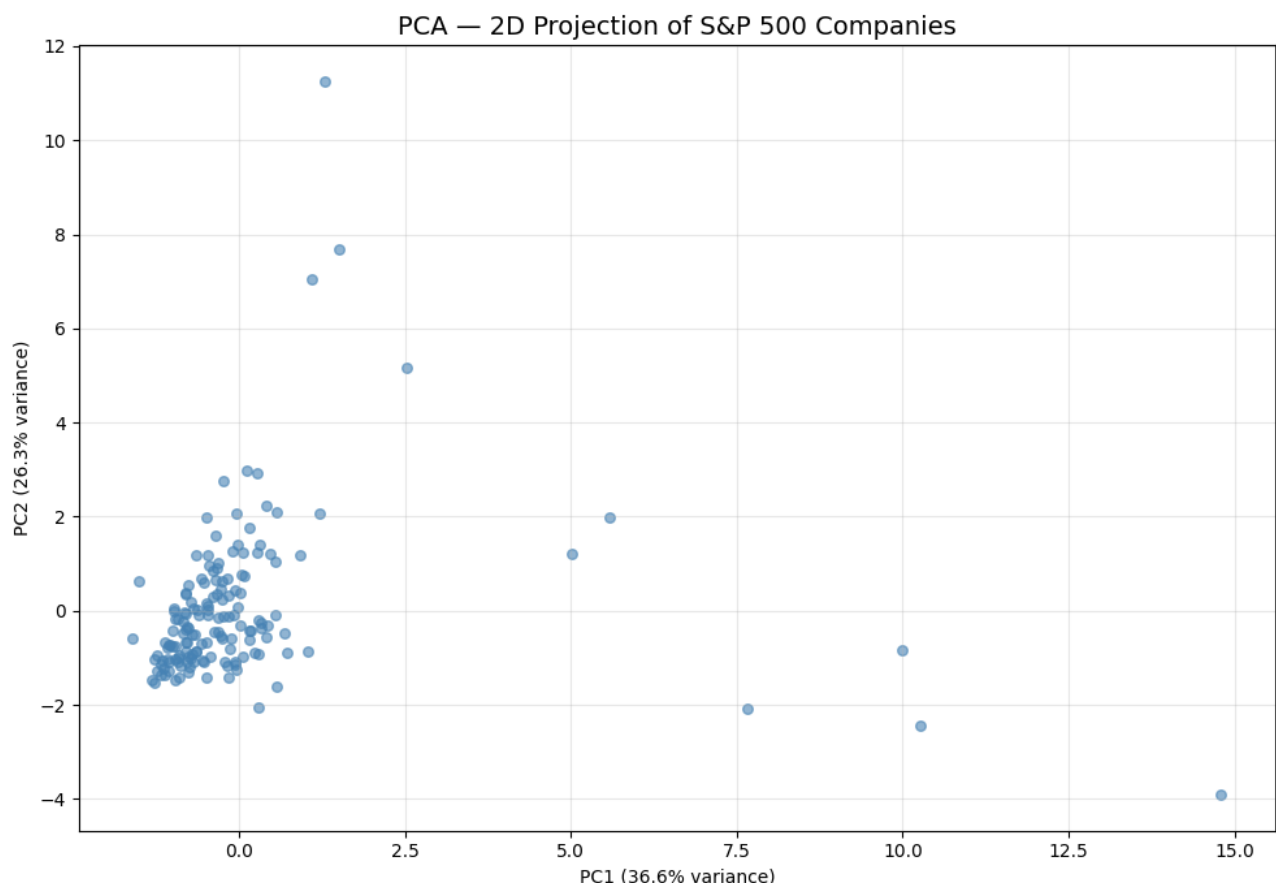


Figure 1: PCA 2D projection of S&P 500 companies. A few outlier companies (mega-cap tech) are clearly separated from the main cluster.

Algorithms and Tuning

K-Means: Tested k = 2 to 10 using the Elbow Method (inertia/WCSS) and Silhouette Scores.

DBSCAN: Systematically varied eps (0.5-5.0) and min_samples (3, 5, 7, 10). Selected the configuration with the highest Silhouette Score among those producing >=2 clusters.

Agglomerative Clustering: Compared three linkage methods (ward, complete, average) across k = 2 to 5, selecting the combination with the highest Silhouette Score. Dendrograms plotted for ward and complete linkage.

3. Results

Clustering Comparison

Algorithm	# Clusters	Silhouette Score	Key Observations
K-Means	2	0.7226	Clean separation; Elbow + Silhouette confirm k=2
DBSCAN	6	0.3622	131 noise points; density-based, identifies outliers
Agglomerative	2	0.8072	Average linkage best; hierarchical structure visible

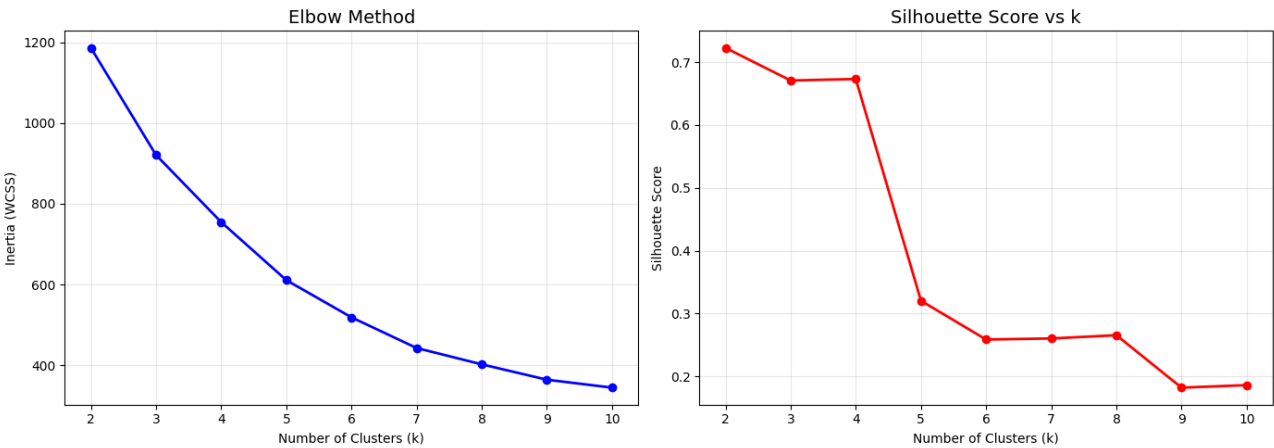


Figure 2: K-Means Elbow plot (left) and Silhouette analysis (right). Both indicate k=2 as optimal.

Clustering Results Comparison (PCA 2D Projection)

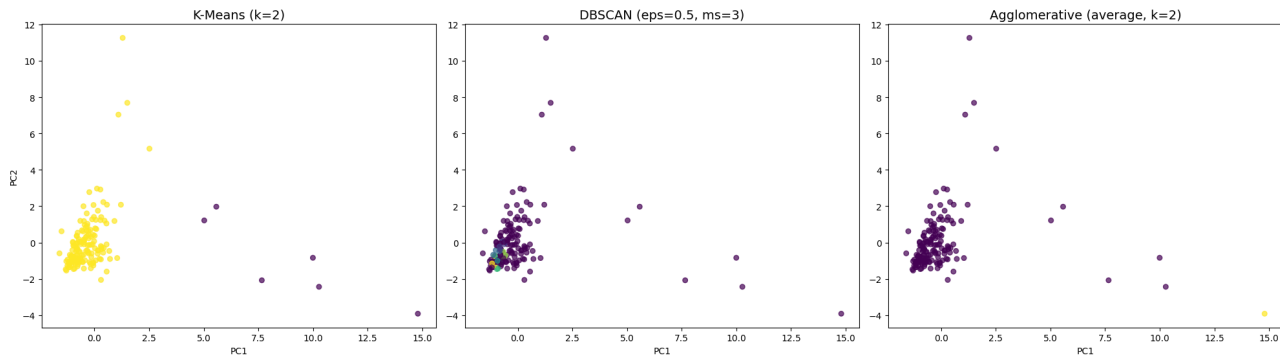


Figure 3: Side-by-side comparison of all three clustering results on PCA 2D projection. K-Means and Agglomerative produce similar 2-cluster solutions; DBSCAN identifies 6 clusters with many noise points.

Linkage Method Comparison (Agglomerative)

n_clusters	Average	Complete	Ward
2	0.807	0.762	0.762
3	0.761	0.738	0.686
4	0.683	0.688	0.688
5	0.684	0.687	0.248

Average linkage with k=2 produced the highest Silhouette Score (0.807).

Key Findings

- The data naturally splits into 2 main clusters: mega-cap companies (Apple, Microsoft, Amazon, etc.) vs. the remaining S&P 500 members, primarily driven by Market Cap, EBITDA, and Weight.
- DBSCAN identified 131 noise points (26% of companies) -- these are companies with unusual combinations of features that don't fit neatly into dense clusters.
- The dendrogram reveals a hierarchical structure: the first major split separates mega-caps, followed by finer divisions by sector and growth characteristics.

4. Conclusion

Recommended Algorithm: Agglomerative Clustering (Average Linkage)

Agglomerative Clustering achieved the highest Silhouette Score (0.8072), producing the cleanest and most interpretable 2-cluster solution. The hierarchical nature also provides the dendrogram, offering insight into sub-group structures within each cluster.

K-Means is a close second (Silhouette = 0.7226) and is more practical for larger datasets due to its computational efficiency. DBSCAN, while useful for identifying outlier companies, produced a lower Silhouette Score due to the non-uniform density of financial data.

For portfolio construction, the 2-cluster solution (mega-cap vs. rest) provides a fundamental diversification axis. Finer-grained clustering (k=3-5) could separate growth vs. value vs. defensive stocks for more nuanced allocation strategies.